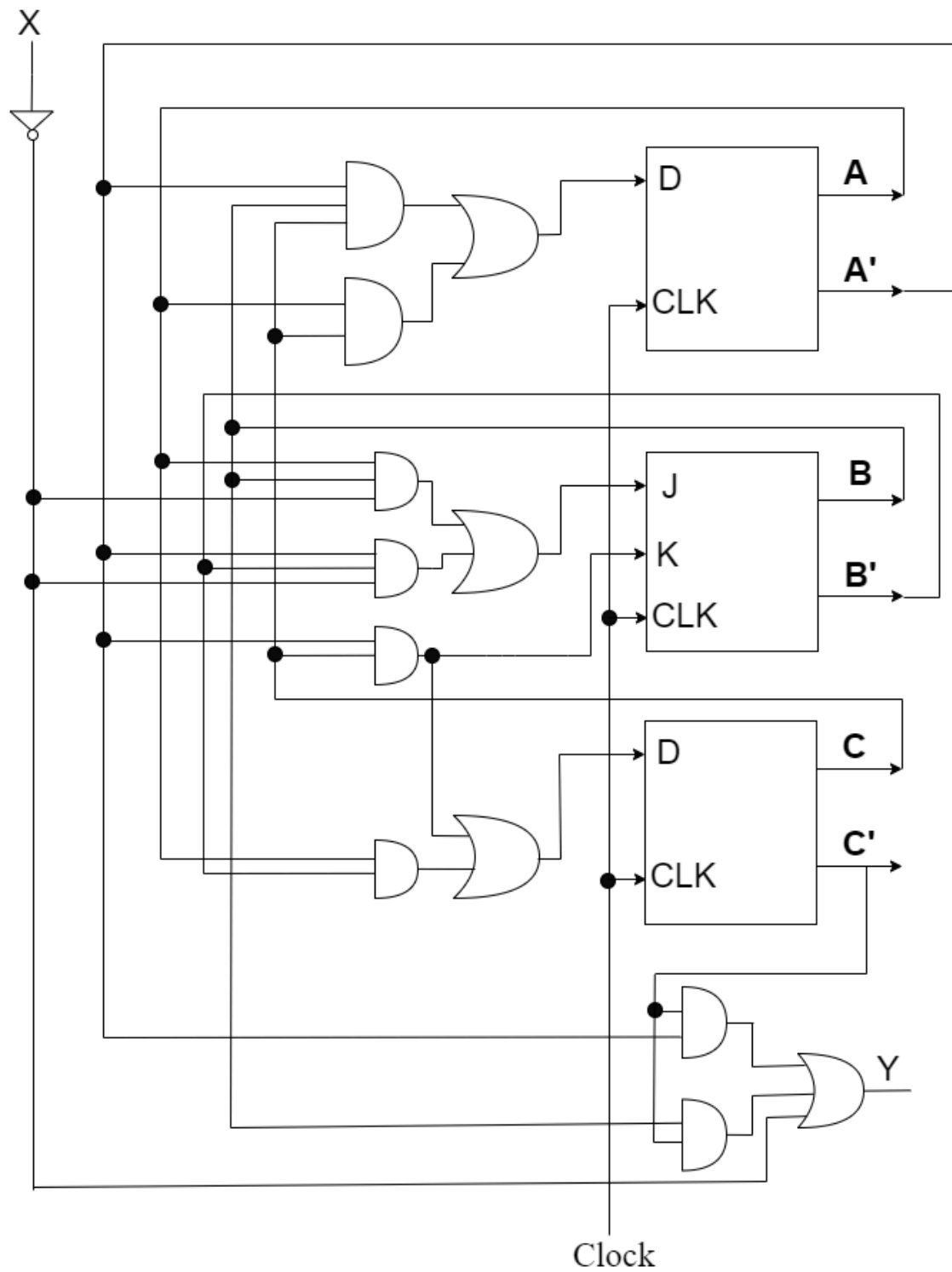


CSE 260
Assignment 4
Total 2X10 = 20

Question 1

Draw the state diagram based upon the circuit diagram given below.



Question 2

You are designing a simple **elevator system** that moves between four floors (0,1,2,3). The elevator should track its current floor using a **2-bit up/down counter**. However, there is a restriction: Transitions directly between the ground floor (0) and the top floor (3), or the top floor (3) and the ground floor (0) are not allowed.

Now, **design** the 2-bit up/down counter using SR flip flop(s).

Note: The circuit is to be designed by treating the unused states (if there is any) as Don't Care conditions.